FFmpeg Converter GUI

Prerequisites & Installation Guide · Version 3.0

Before you can use FFmpeg Converter GUI, two things need to be in place: the FFmpeg binary on your system, and (recommended) a supported GPU for hardware-accelerated encoding. As of version 3.0, GPUs from **NVIDIA, AMD, and Intel** are all supported.

1. FFmpeg 8.0 or newer (required)

FFmpeg Converter GUI is a graphical front-end for FFmpeg. The actual video conversion is done by the FFmpeg command-line tool, which must be installed separately.

Why version 8.0 or newer? The GUI uses modern FFmpeg syntax such as `-init_hw_device cuda:N` (NVIDIA), `d3d11va:N` (AMD), `qsv:N` (Intel) for GPU device selection, and `-fps_mode cfr` for Mediathek-Safe mode. These options are not available in older FFmpeg versions and the GUI will refuse to start the encoding process if FFmpeg is too old.

Installation:

- Download a recent build from `https://ffmpeg.org/download.html` (Windows users: the “gyan.dev” or “BtbN” builds are recommended — they include NVENC, AMF, and QSV).
- Extract the archive and either add the `bin/` folder to your system PATH, or copy `ffmpeg.exe` and `ffprobe.exe` into the same folder as the GUI executable.
- Verify the installation by opening a terminal and running `ffmpeg -version` — the first line must report version 8.0 or higher.

2. Supported GPU for hardware encoding (recommended)

The GUI supports hardware encoders from three vendors, each with its own dedicated encoding silicon. Hardware encoding is dramatically faster than software (CPU-only) encoding — typically by a factor of 5x to 20x, depending on codec, resolution, and the GPU generation. For 4K material, hardware encoding is essentially mandatory.

Without a supported GPU the GUI still works, but encoding falls back to the CPU and even short clips can take many minutes per minute of video.

Recommended hardware per vendor:

- **NVIDIA NVENC** — AV1: GeForce RTX 40-series or newer (Ada Lovelace; first NVENC generation with AV1). H.265 / H.264: any GeForce GTX 10-series or newer. Detection via `nvidia-smi`; current NVIDIA driver required.
- **AMD AMF** *(new in v3.0)* — AV1: Radeon RX 7000-series or newer (RDNA 3). H.265 / H.264: any Radeon RX 400-series or newer. Detection via Windows WMI or Linux `lspci`; current Adrenalin / Mesa driver required.
- **Intel Quick Sync** *(new in v3.0)* — AV1: Arc discrete GPUs or 13th-gen Core integrated graphics. H.265 / H.264: any 6th-gen Core (Skylake) or newer. Detection same as AMD.
- **Software fallback** — SVT-AV1, x265, x264, libvpx-vp9 always available regardless of GPU. Slower, but produces excellent quality at the cost of CPU time.

3. Quick checklist

Step	How to verify
FFmpeg 8.0+ in PATH	Run `ffmpeg -version` in a terminal
ffprobe 8.0+ available	Run `ffprobe -version` in a terminal
Hardware encoders compiled in	Run `ffmpeg -encoders` and look for `nvenc` / `_amf` / `_qsv` entries
GPU driver installed	NVIDIA: `nvidia-smi` · AMD/Intel: GPU appears in Device Manager / `lspci`
GPU recognised by GUI	Open the app: the “Encoder” dropdown lists your GPU(s). Only encoders matching your selected vendor are shown.